

CURRICULUM VITAE

DR CARLOS SÁNCHEZ MUÑOZ

PERSONAL DATA

Birth 14/09/1988, Ciudad Real (Spain)

Languages Spanish (native), English (fluent),
Japanese (Basic)

Job address Instituto de Física Fundamental, Calle
Serrano, 113, 28006, Madrid

Phone (+34) 686655120

Email carlos.sanchez@iff.csic.es,
carlossmwolff@gmail.com

Web <https://carlossanchezmunoz.weebly.com/>

Online profiles: [Google Scholar](#), [Researchgate](#), [Publons](#).

ResearcherID Q-1509-2017



RESEARCH FIELDS

Quantum optics, quantum technologies, light-matter interactions, quantum metrology, generation of quantum radiation, open quantum systems, quantum nanomechanics.

CURRENT POSITION

2024– **Permanent Researcher (Científico Titular)**
Institute of Fundamental Physics (IFF), CSIC, Madrid.

PREVIOUS RESEARCH EXPERIENCE

2020–2023 **Postdoctoral Researcher, La Caixa Junior Leader Fellow.**
IFIMAC, Universidad Autónoma de Madrid

2018–2020 **Postdoctoral Researcher, Marie Curie Fellow.**
University of Oxford, UK.

2017–2018 **RIKEN, Japan.** Postdoctoral Researcher, JSPS Fellow.

EDUCATION

2013–2016 **PhD in Condensed Matter Physics and Nanotechnology.**

	Universidad Autónoma de Madrid, Spain. Highest distinction “Sobresaliente cum Laude”. Premio Extraordinario de Doctorado (“Extraordinary Award”) from Universidad Autónoma de Madrid. Thesis Title: <i>Generation of Nonclassical States of light from Nanostructures</i>. Supervisors: Carlos Tejedor and Fabrice P. Laussy.
2011–2012	MSc in Condensed Matter Physics and Nanotechnology. Universidad Autónoma de Madrid, Spain.
2006–2011	Five-year degree in Physics. Universidad Autónoma de Madrid, Spain.

GRANTS AND AWARDS

2021	SNSF Scientific Exchanges Grant. Swiss National Science Foundation. Grant to cover international research visits to Swiss Research Institutions.
2020	La Caixa Junior Leader Postdoctoral Fellowship. 3-years postdoctoral grant covering fellow salary and 38k€ of research costs per year.
2019	Juan de la Cierva-Incorporación 2018. Spanish Ministry of Science (declined)
2018	Extraordinary Thesis Award. Universidad Autónoma de Madrid
2017	GEFES Award 2017 Best Theoretical Thesis Best theoretical PhD thesis in Condensed Matter Physics, awarded by the Condensed Matter Division of the Spanish Royal Society of Physics.
2017	Marie Skłodowska-Curie Individual Fellowships. Call: H2020-MSCA-IF-2016. Proposal acronym: QUSON.
2017	JSPS Postdoctoral Fellowship for Research in Japan (Short-term).. Japan Society for the Promotion of Science.
2014-2015	FPI Mobility Grant. Grant awarded among FPI fellows to fund short-term stays in foreign institutions during the PhD. Grant awarded twice to visit Max-Planck-Institute for Quantum Optics (Garching, Germany) and University of Sussex (Brighton, United Kingdom).
2013	FPI predoctoral Grant. Four-year grant awarded by Spanish Ministry of Economy and Competitiveness for PhD studies.
2011-2012	Grant for the beginning of Postgraduate programmes from Universidad Autónoma de Madrid. Grant devoted to support research work during the first stages of the researcher career at Universidad Autónoma de Madrid.
2010–2011	Collaboration grant. Undergraduate research scholarship awarded by Spanish Ministerio de Educación y Ciencia.

INTERNATIONAL RESEARCH STAYS AND VISITS

04/2023	RIKEN, Wako-shi, Japan One-month visit.
---------	---

05/2022	EPFL, Lausanne, Switzerland. Three-month visit.
01/2022	Chalmers University of Technology, Gothenburg, Sweden. Three-week visit.
01/2020	RIKEN, Wako-shi, Japan. One week visit.
01/2019	RIKEN, Wako-shi, Japan. One week visit.
08/2018	Institute for Quantum Optics and Quantum Information (IQOQI), Innsbruck, Austria. Two-days visit.
10/2015	Max-Planck-Institute for Quantum Optics, Garching, Germany. One week visit.
04/2015–07/2015	University of Sussex, Brighton, United Kingdom. Three month stay at the Atomic, Molecular and Optical Physics Research Group under the supervision of Dr. Diego Porras.
09/2014–12/2014	Max-Planck-Institute for Quantum Optics, Garching, Germany. Three month stay at the Theory Division under the supervision of Prof. Dr. J. Ignacio Cirac. Stay funded by the Spanish Ministry of Economy and Competitiveness.

PUBLICATIONS

41 peer-reviewed publications, 11 papers as first author, 8 papers as last author. These include 1 Nature Photonics (1 first author), 1 Nature Communications (1 first author), 1 Nature Materials, 1 Science Advances, 14 Physical Review Letters (3 first author, 2 last author), 1 PRX Quantum (1 last author). **h-index: 24.**
Citations: 2013 (Google Scholar, as of January 10, 2025).

1. *Frequency-resolved Purcell effect for the dissipative generation of steady-state entanglement.*
A. Vivas-Viaña, D. Martín-Cano, **C. Sánchez Muñoz**
[Physical Review Letters 133 \(17\), 173601 \(2024\)](#)
2. *Dissipative stabilization of maximal entanglement between nonidentical emitters via two-photon excitation.*
A. Vivas-Viaña, D. Martín-Cano, **C. Sánchez Muñoz**
[Physical Review Research 6 \(4\), 043051 \(2024\)](#)
3. *Quantum amplification and simulation of strong and ultrastrong coupling of light and matter.*
W. Qin, A. Frisk Kockum, **C. Sánchez Muñoz**, A. Miranowicz, F. Nori
[Physics Reports 1078, 1-59 \(2024\)](#)
4. *Parameter estimation from quantum-jump data using neural networks.*
E. Rinaldi, M. González Lastre, S. García Herreros, S. Ahmed, M. Khanahmadi, F. Nori, **C. Sánchez Muñoz**
[Quantum Science and Technology 9 \(3\), 035018 \(2024\)](#)
5. *Strongly coupled spin waves and surface acoustic waves at room temperature.*
Y. Hwang, J. Puebla, K. Kondou, C. Gonzalez-Ballester, H. Ishiki, **C. Sánchez Muñoz**, L. Liao, F.A. Chen, W. Luo, S. Maekawa, Y. Otani
[Physical Review Letters 132 \(5\), 056704 \(2024\)](#)
6. *Single-photon source over the terahertz regime.*
C. Groiseau, A. I Fernández-Domínguez, D. Martín-Cano, **C. Sánchez Muñoz**
[PRX Quantum 5 \(1\), 010312 \(2024\)](#)

7. *Resolving nonclassical magnon composition of a magnetic ground state via a qubit.*
A. E Römling, A. Vivas-Viaña, **C. Sánchez Muñoz**, A. Kamra
[Physical Review Letters 131 \(14\), 143602 \(2023\)](#)
8. *Nonlinear acoustic spin pumping caused by temperature-dependent frequency shifts of surface acoustic waves.*
Y. Hwang, J. Puebla, K. Kondou, **C. Sánchez Muñoz**, Y. Otani
[Journal of the Physical Society of Japan 92 \(9\), 094702 \(2023\)](#)
9. *Nonlinear interferometry for quantum-enhanced measurements of multiphoton absorption.*
S. Panahiyan, **C. Sánchez Muñoz**, M. V Chekhova, F. Schlawin
[Physical Review Letters 130 \(20\), 203604 \(2023\)](#)
10. *On the generality of symmetry breaking and dissipative freezing in quantum trajectories.*
J. Tindall, D. Jaksch, **C. Sánchez Muñoz**
[SciPost Physics Core 6 \(1\), 004 \(2023\)](#)
11. *Interaction between giant atoms in a one-dimensional structured environment.*
A. Soro, **C. Sánchez Muñoz**, A. Frisk Kockum
[Physical Review A 107 \(1\), 013710 \(2023\)](#)
12. *Spontaneous scattering of Raman photons from cavity-QED systems in the ultrastrong coupling regime.*
V. Macrì, A. Mercurio, F. Nori, S. Savasta, **C. Sánchez Muñoz**
[Physical Review Letters 129 \(27\), 273602 \(2022\)](#)
13. *Two-photon-absorption measurements in the presence of single-photon losses.*
S. Panahiyan, **C. Sánchez Muñoz**, M. V Chekhova, F. Schlawin
[Physical Review A 106 \(4\), 043706 \(2022\)](#)
14. *Unconventional mechanism of virtual-state population through dissipation.*
A. Vivas-Viaña, A. González-Tudela, **C. Sánchez Muñoz**
[Physical Review A 106, 012217 \(2022\)](#)
15. *Squeezed lasing.*
C. Sánchez Muñoz, D. Jaksch
[Physical Review Letters 127 \(18\), 183603 \(2021\)](#)
16. *Quantum state tomography with conditional generative adversarial networks.*
S. Ahmed, **C. Sánchez Muñoz**, F. Nori, A. Frisk Kockum
[Physical review letters 127 \(14\), 140502 \(2021\)](#)
17. *Quantum metrology of two-photon absorption.*
C. Sánchez Muñoz, G. Frascella, F. Schlawin
[Physical Review Research 3, 033250 \(2021\)](#)
18. *Classification and reconstruction of optical quantum states with deep neural networks.*
S. Ahmed, **C. Sánchez Muñoz**, F. Nori, A. Frisk Kockum
[Physical Review Research 3 \(3\), 033278 \(2021\)](#)
19. *Two-photon resonance fluorescence of two interacting non-identical quantum emitters.*
A. Vivas-Viaña, **C. Sánchez Muñoz**
[Physical Review Research 3, 033136 \(2021\)](#)
20. *Simulating ultrastrong-coupling processes breaking parity conservation in Jaynes-Cummings systems.*
C. Sánchez Muñoz, A. Frisk Kockum, A. Miranowicz, F. Nori
[Physical Review A 102 \(3\), 033716 \(2020\)](#)

21. *Photon correlation spectroscopy as a witness for quantum coherence.*
C. Sánchez Muñoz, F. Schlawin
[Physical Review Letters 124 \(20\), 203601 \(2020\)](#)
22. *Quantum synchronisation enabled by dynamical symmetries and dissipation.*
J. Tindall, **C. Sánchez Muñoz**, B. Buča, D. Jaksch
[New Journal of Physics 22 \(1\), 013026 \(2020\)](#)
23. *Symmetries and conservation laws in quantum trajectories: Dissipative freezing.*
C. Sánchez Muñoz, B. Buča, J. Tindall, A. González-Tudela, D. Jaksch, D. Porras
[Physical Review A 100 \(4\), 042113 \(2019\)](#)
24. *Engineering and harnessing giant atoms in high-dimensional baths: a proposal for implementation with cold atoms.*
A. González-Tudela, **C. Sánchez Muñoz**, J. Ignacio Cirac
[Physical review letters 122 \(20\), 203603 \(2019\)](#)
25. *Hybrid systems for the generation of nonclassical mechanical states via quadratic interactions.*
C. Sánchez Muñoz, A. Lara, J. Puebla, F. Nori
[Physical review letters 121 \(12\), 123604 \(2018\)](#)
26. *Resolution of superluminal signalling in non-perturbative cavity quantum electrodynamics.*
C. Sánchez Muñoz, F. Nori, S. De Liberato
[Nature communications 9 \(1\), 1924 \(2018\)](#)
27. *First observation of the quantized exciton-polariton field and effect of interactions on a single polariton.*
Á. Cuevas, J. Camilo López Carreño, B. Silva, M. De Giorgi, D. G Suárez-Forero, **C. Sánchez Muñoz**,
A. Fieramosca, F. Cardano, L. Marrucci, V. Tasco, G. Biasiol, E. Del Valle, L. Dominici, D. Ballarini, G.
Gigli, P. Mataloni, F. P Laussy, F. Sciarrino, D. Sanvitto
[Science Advances 4 \(4\), eaao6814 \(2018\)](#)
28. *Topological order and thermal equilibrium in polariton condensates.*
D. Caputo, D. Ballarini, G. Dagvadorj, **C. Sánchez Muñoz**, M. De Giorgi, L. Dominici, K. West, L. N
Pfeiffer, G. Gigli, F. P Laussy, M. H Szymańska, D. Sanvitto
[Nature materials 17 \(2\), 145-151 \(2018\)](#)
29. *Filtering multiphoton emission from state-of-the-art cavity quantum electrodynamics.*
C. Sánchez Muñoz, F. P Laussy, E. Del Valle, C. Tejedor, A. González-Tudela
[Optica 5 \(1\), 14-26 \(2018\)](#)
30. *Macroscopic Two-Dimensional Polariton Condensates.*
D. Ballarini, D. Caputo, **C. Sánchez Muñoz**, M. De Giorgi, L. Dominici, M. H Szymańska, K. West, L.
N Pfeiffer, G. Gigli, F. P Laussy, D. Sanvitto
[Physical review letters 118 \(21\), 215301 \(2017\)](#)
31. *Excitation with quantum light. II. Exciting a two-level system.*
J.C. López Carreño, **C. Sánchez Muñoz**, E. Del Valle, F.P. Laussy
[Physical Review A 94 \(6\), 063826 \(2016\)](#)
32. *The colored hanbury brown–twiss effect.*
B. Silva, **C. Sánchez Muñoz**, D. Ballarini, A. González-Tudela, M. De Giorgi, G. Gigli, K. West, L.
Pfeiffer, E. Del Valle, D. Sanvitto, F.P. Laussy
[Scientific reports 6 \(1\), 37980 \(2016\)](#)
33. *Deterministic down-converter and continuous photon-pair source within the bad-cavity limit.*
Y. Chang, A. González-Tudela, C. Sanchez Munoz, C. Navarrete-Benlloch, T. Shi
[Physical review letters 117 \(20\), 203602 \(2016\)](#)

34. *Enhanced two-photon emission from a dressed biexciton.*
C. Sánchez Muñoz, F. P Laussy, C. Tejedor, E. Del Valle
[New Journal of Physics 17 \(12\), 123021 \(2015\)](#)
35. *Exciting polaritons with quantum light.*
J.C. López Carreño, **C. Sánchez Muñoz**, D. Sanvitto, E. Del Valle, F. P Laussy
[Physical Review Letters 115 \(19\), 196402 \(2015\)](#)
36. *Degenerate parametric oscillation in quantum membrane optomechanics.*
M. Benito, C. Sánchez Muñoz, C. Navarrete-Benlloch
[Physical Review A 93, 023846 \(2016\) \(2015\)](#)
37. *Coherent generation of nonclassical light on chip via detuned photon blockade.*
K. Müller, A. Rundquist, K. A Fischer, T. Sarmiento, K. G Lagoudakis, Y. A Kelaita, C. Sanchez Muñoz, E. Del Valle, F. P Laussy, J. Vučković
[Physical review letters 114 \(23\), 233601 \(2015\)](#)
38. *Violation of classical inequalities by photon frequency filtering.*
C. Sánchez Muñoz, E. Del Valle, C. Tejedor, F. P Laussy
[Physical Review A 90 \(5\), 052111 \(2014\)](#)
39. *Emitters of N-photon bundles.*
C. Sánchez Muñoz, E. Del Valle, A. González Tudela, K. Müller, S. Lichtmannecker, M. Kaniber, C. Tejedor, J.J. Finley, F.P. Laussy
[Nature photonics 8 \(7\), 550-555 \(2014\)](#)
40. *Bichromatic dressing of a quantum dot detected by a remote second quantum dot.*
M. Maragkou, C. Sánchez-Muñoz, S. Lazić, E. Chernysheva, H. Piet Van Der Meulen, A. González-Tudela, C. Tejedor, L.J. Martínez, I. Prieto, P.A. Postigo, J.M. Calleja
[Physical Review B—Condensed Matter and Materials Physics 88 \(7\), 075309 \(2013\)](#)
41. *Plasmon-polariton emission from a coherently -excited quantum dot near a metal interface.*
C. Sanchez-Munoz, A. Gonzalez-Tudela, C. Tejedor
[Physical Review B—Condensed Matter and Materials Physics 85 \(12\), 125301 \(2012\)](#)

PREPRINTS

1. *Photon Antibunching in Single-Molecule Vibrational Sum-Frequency Generation.*
F. Moradi Kalarde, C. Sanchez Munoz, J. Feist, C. Galland
[arXiv: 2409.05124 \(2024\)](#)
2. *Entanglement of photonic modes from a continuously driven two-level system.*
J. Yang, I. Strandberg, A. Vivas-Viana, A. Gaikwad, C. Castillo-Moreno, A. Frisk Kockum, M. Asad Ullah, C. Sanchez Munoz, A. Martin Eriksson, S. Gasparinetti
[arXiv: 2407.07991 \(2024\)](#)

RESEARCH PROJECTS

Participation in 9 Research Projects, being PI of 4 of them. Since the beginning of my postdoctoral stage in 2017, I have raised a total of 0.7M€ in competitive, individual funding schemes, and as a PI, I have been responsible of managing Research budget of more than 215k€.

PRINCIPAL INVESTIGATOR (PI)

1. **Sensores con Correlaciones Cuánticas en Sistemas Luz Materia Abiertos (SEQUOIA) (PID2023-149969NA-I00).**

Funding agency: Ministerio de Ciencia e Innovación, Proyecto de Generación de Conocimiento 2023.
Duration: 09/2024-09/2027. Funding: ~75k€. PIs: Carlos Sánchez Muñoz.

2. **La Caixa Junior Leader: Multiphoton sources from single quantum emitters.**

Funding agency: Obra Social Fundación la Caixa. Duration: 01/08/2020-31/07/2023. Funding: 305.100 €. PIs: Carlos Sánchez Muñoz.

3. **Marie Curie Individual Fellowship (QUSON).**

Funding agency: European Union, Program Horizon 2020. University of Oxford, UK. Duration: 04/06/2018-04/06/2020. Funding: 183.454,8 €. PIs: Carlos Sánchez Muñoz.

4. **JSPS Short-term Fellowship Japanese Society for the Promotion of Science.**

Funding agency: Japanese Society for the Promotion of Science. RIKEN, Japan. Duration: 01/10/2017-30/04/2018. Funding: ~43k€ €. PIs: Carlos Sánchez Muñoz.

WORK TEAM

1. **Ingeniería Cuántica de Luz y Materia en la Nanoescala (PID2021-126964OB-I00).**

Funding agency: Ministerio de Ciencia e Innovación, convocatoria 2021 de Proyectos de Generación de Conocimiento 2021. Duration: 09/2022-09/2025. Funding: ~150k€. PIs: Antonio Fernández Domínguez, Diego Martín Cano.
Role: Member of the work team.

2. **Nanofotónica para Computación Cuántica (NanoQuCo).**

Funding agency: Comunidad de Madrid, Y2020/TCS-6545. Duration: 07/2022-07-2025. Funding: 657.800€. PIs: Francisco José García Vidal and Juan José García Ripoll.
Role: Member of the work team, collaboration in the project design.

3. **Frontiers in Quantum Simulation.**

Funding agency: Ministerio Ciencia, Innovación y Universidades. PI: Diego Porras, IFF, CSIC, Madrid.
Role: Part of the work team.

4. **Quantum Optics in nanostructured semiconductors (QOINS).**

Funding agency: MINECO. PI: Luis Viña. Universidad Autónoma de Madrid. Duration: 2015/2017.
Role: Member of the work team.

5. **Nanoestructuras para óptica cuántica (NANOQUO).**

Funding agency: MINECO. PI: Carlos Tejedor. Funding: 466.942€ . Universidad Autónoma de Madrid.
Role: Member of the work team.

TALKS AND PRESENTATIONS

I have attended 26 conferences, where I have given 21 talks, 9 of them invited talks.

I have given 16 invited seminars.

CONFERENCES

2024

1. Multiphotonics, Garching, Germany, July 2024.

Invited Talk: *Strong driving of quantum emitters: routes towards entanglement and multiphoton generation*

2023

2. CMD 30 - FisMat 2023, Milan, Italy, September 2023.
Invited Talk: *Spontaneous Scattering of Raman photons from cQED systems in the Ultrastrong Coupling Regime*
3. Central European Workshop on Quantum Optics, Milan, Italy, July 2023
Talk: *Dissipative generation of high stationary entanglement between non-identical quantum emitters*
4. Conferencia Española de Nanofotónica CEN 2023, Zaragoza, Spain, June 2023
Invited Talk: *Strongly driven quantum emitters for quantum applications*
5. Quantum Matter 2023, Madrid, Spain, May 2023
Talk: *Machine Learning for Parameter Estimation from Continuously-Monitored Quantum Systems.*

2022

6. FRIAS Junior Researcher Conference - Quantum Control of Complex Systems, Freiburg, Germany. September 2022.
Invited Talk: *Squeezed Lasing.*
7. Molecular Polaritonics 2022. Straubing, Germany. September 2022.
Invited Talk: *Squeezed Lasing and Quantum Metrology of Two-photon absorption.*

2021

8. ICE-6, 6th Quantum Information Conference in Spain. Online Conference. May 2021.
Talk: *Squeezed Lasing.*

2020

9. Physics of Light-Matter Coupling in Nanostructures 2020 (PLMCN 2020). Clermont-Ferrand, France. October 2020.
Talk: *Ultrastrong-coupling effects induced by a single classical drive in Jaynes-Cummings-type systems.*
10. Sensing with Quantum Light 2020. September 2020.
Talk: *Squeezed Lasing.*
11. 10th International Conference on Spontaneous Coherence in Excitonic Systems (ICSCE10). Melbourne, Australia. January 2020.
Invited talk: *Ultrastrong coupling in cavity QED: Exotic phenomena and their simulation.*

2019

12. Quantum light for investigating complex molecules and materials, Royal Society Meeting, UK, November 2019.
Poster: *Emitters of N-photon bundles..*
13. IoP Lectures.
University of Wolverhampton, UK, October 2019.
Invited talk: *The tiniest possible sound.*
14. DesOEQ (Designing out-of-equilibrium many-body quantum systems)-QSUM (Quantum Science with Ultracold Molecules) Meeting.
Oxford, UK, September 2019.
Talk: *Quantum synchronization enabled by dynamical symmetries and dissipation.*

15. Terametanano 4: International Conference on Terahertz Emission, Metamaterials and Nanophotonics. Lecce, Italy. May 2019.
Invited talk: Resolution of superluminal signalling in the ultrastrong coupling regime of cavity QED.
2018
16. XGEFES 2018 Meeting (División de Física de la Materia Condensada de la Real Sociedad Española de Física). Valencia, España. Enero 2018.
Invited talk: *Generation of Non-classical States of Light*
17. 2nd International Symposium on Dynamics in Artificial Quantum Systems (DAQS2018). Tokyo, Japón. January 2018.
 Poster: *Multi-mode Quantum Rabi Model and Superluminal Signalling*
2017
18. International School and Symposium on Nanoscale Transport and phoTonics (ISNTT2017). Atsugi, Japan. November 2017.
 Poster: *Multi-mode Quantum Rabi Model and Superluminal Signalling*
2016
19. 33rd International Conference On the Physics of Semiconductors (ICPS 2016). Beijing, China. August 2016.
 Talk: *Bundles of entangled photons emitted from quantum dots in microcavities.*
20. Physics of Light-Matter Coupling in Nanostructures 2016 (PLMCN17). Nara, Japan. March 2016.
 Talk: *Emitters of bundles of entangled photons.*
2015
21. XI Reunión Nacional de Óptica. Salamanca, Spain. September 2015.
 Talk: *Degenerate parametric oscillation in membrane cavity optomechanics.*
22. International Conference on Quantum Technologies (ICQT 2015). Moscow, Russia. July 2015.
 Poster: *Photon bundles: A new type of light*
2014
23. 32nd International Conference On the Physics of Semiconductors (ICPS 2014). Austin, Texas. August 2014.
 Talk: *Emitters of N-photon bundles.*
24. Physics of Light-Matter Coupling in Nanostructures 2014 (PLMCN 2014). Montpellier, France. June 2014.
 Talk: *Quantum optics with photon bundles.*
2013
25. XVI Young Researchers Meeting, Instituto Nicolás Cabrera. Miraflores, Spain. December 2013.
 Talk: *Emitters of N-photon states.*

26. International Conference on Optics of Excitons in Confined Systems (OECS 13).
Rome, Italy. September 2013.
Poster: *An emitter of N -photon states.*

SEMINARS

2022

1. *Squeezed Lasing*
April 2022, PhD and Young Scientists Quantum Technologies Network (PYSQT) .Invited Seminar,
Online Event. [Link to the talk.](#)
2. *Squeezed Lasing*
January 2022, Wallenberg Centre for Quantum Technology, Chalmers University of Technology,
Gothenburg (Sweden). [Link to online announcement.](#)

2020

3. *Generation and application of quantum states of light*
November 2020, Department of Theoretical Condensed Matter Physics, Informal Seminars, UAM,
Madrid (Spain)
4. *Squeezed Lasing*
October 2020, University of Oxford (UK)
5. *Squeezed Lasing*
June 2020, Institute for Fundamental Physics, CSIC, Madrid (Spain)
6. *Squeezed Lasing*
June 2020, Wilczek Quantum Center Seminars, Shanghai (China)

2019

7. *Symmetries and Conservation Laws in Quantum Trajectories: Dissipative Freezing*
July 2019, Institute for Fundamental Physics, CSIC, Madrid (Spain)

2018

8. *Physics of strongly-driven cavity QED systems.*
October 2018. Atomic and Laser Physics Seminar, University of Oxford, UK.
9. *Two-phonon interactions for the generation of mechanical and spin cat states.*
August 2018.
Institute for Quantum Optics and Quantum Information (IQOQI), Innsbruck, Austria.

2017

10. *Exploiting quantum correlations in the frequency domain.*
February 2017.
Informal Workshop: EPR steering, temporal steering, and correlations in quantum theory. The Quantum
Condensed Matter Research Group, RIKEN, Wako-shi, Saitama, Japan.

2015

11. *Degenerate Parametric Amplification in membrane cavity quantum optomechanics.*
December 2015.
Department of Theoretical Condensed Matter Physics Seminar, Universidad Autónoma de Madrid.

12. *Quantum gates with polaritons.*
April 2015.
Polaflow Annual Meeting, Madrid
13. *Secret photon affairs.*
February 2015.
Department of Theoretical Condensed Matter Physics Seminar, Universidad Autónoma de Madrid.

2014

14. *Frequency-resolved photon correlations: discovering sources of quantum light*
October 2014.
Theory Seminar, Max-Planck-Institute for Quantum Optics, Garching, Germany.

2013

15. *N-photon physics with cQED systems; violation of the Cauchy-Schwartz inequality.*
September 2013. Polaflow Annual Meeting, Lecce, Italy.
16. *Towards a Two-Photon laser; Strategies with quantum dots in a cavity.*
April 2013. Schottky Seminars, Walter Schottky Institut, Garching, Germany.

THESES DIRECTION

PHD THESIS

- | | |
|-----------|--|
| 2025- | Rodrigo Grande de Diego, PhD Thesis (ongoing). Funded personally by Proyecto de Generación de Conocimiento 2023 SEQUOIA (FPI Grant). |
| 2020-2025 | Alejandro Vivas Viaña, PhD Thesis. Title: <i>Dynamics of light-matter dressing in Collective Quantum Optics</i> . Universidad Autónoma de Madrid. Funded personally by La Caixa Junior Leader Project. |

MASTER THESIS

- | | |
|-----------|---|
| 2024-2025 | Rodrigo Grande de Diego, MsC Project. Title: <i>Squeezed lasing with cold atoms</i> . Universidad de Granada. |
| 2020-2021 | Javier Jorge Gallo, MsC Project. Title: <i>Cascaded quantum-optical systems</i> . Universidad Autónoma de Madrid. |
| 2019-2020 | Alexander Saoulis, MsC Project. Title: <i>Non-linear quantum optical processes enabled by classical driving</i> . University of Oxford. |

TFGs (END OF DEGREE THESES)

- | | |
|-----------|--|
| 2022-2023 | Rodrigo Grande de Diego, TFG Project. Title: <i>Chaos and quantum-classical correspondence in an open system</i> . Universidad Autónoma de Madrid. |
| 2021-2022 | Sergio García Herreros, TFG Project. Title: <i>Neural Networks for Quantum Imaging</i> . Universidad Autónoma de Madrid. |
| 2020-2021 | Manuel Eduardo González Lastre, TFG Project. Title: <i>Redes Neuronales para Metrología Cuántica</i> . Universidad Autónoma de Madrid. |

2020-2021	Miguel Martínez Pilar, TFG Project. Title: <i>Desarrollo de láseres no clásicos</i> . Universidad Autónoma de Madrid.
2019-2020	Alexander Saoulis, MsC Project. Title: <i>Non-linear quantum optical processes enabled by classical driving</i> . University of Oxford.

TEACHING

- More than **190 hours of certified teaching** at Universidad Autónoma de Madrid and University of Oxford.
- **Lecturership in Physics at Corpus Christi College**, University of Oxford, during the academic years 2018/19 and 2019/20.
- **ANECA Contradado Doctor** qualification.

2022-2023	Quantum Optics (Máster Universitario en Física de la Materia Condensada y de los Sistemas Biológicos). Universidad Autónoma de Madrid.
2021-2022	Laboratory of Electronics (CS Major). Universidad Autónoma de Madrid.
2021-2022	General Physics (Chemistry Major). Universidad Autónoma de Madrid.
2020-2021	Laboratory of Electronics (CS Major). Universidad Autónoma de Madrid.
2020-2021	General Physics (Chemistry Major). Universidad Autónoma de Madrid.
2019-2020	Thermal Physics. Corpus Christi College, University of Oxford.
2019	Quantum Mechanics. Corpus Christi College, University of Oxford.
2018	Laboratory Demonstrator (Electromagnetism). University of Oxford.
2012–2016	Laboratory Demonstrator (General Physics). Universidad Autónoma de Madrid.

OUTREACH

2023	Ciencia y palomitas – Introducción a la luz. In this activity, high school students watched our video, "Seriously, What Exactly Is Light?" After some popcorn and a few questions, Carlos Antón Solanas and Carlos Sanchez Muñoz demonstrated to the students how light behaves and how its properties can help us uncover the nature of things..
2022	Artworks for outreach. I won the competition <i>Quantum Visions</i> , an quantum-themed art context organized by UK National Quantum Technologies Programme (UKNQTP), aiming to promote quantum sciences through artworks. <i>A Brumal Coherence Dream</i> was the name of the winning piece. Link to the competition results .
2022	FUTURO Cuántico , online event. Participation on an online encounter with industry professionals for discussions around quantum technologies. Participated with the talk: <i>Quantum Metrology</i> . Link to the event site .
2021	Quantum Fracture: Now, seriously, what is light?. Participation in the making of an outreach video in the most famous science Youtube channel in Spanish (Quantum Fracture, 2.96M subscribers as of December 2021). The video has 1.6M views as of July 2022. Link to the video .

- 2019 **European Researcher's Night**, London , UK.
EU-Pub at the Natural History Museum: Activity akin to a science cafe, where European researchers discuss about science with the general public in an informal setting.
- 2018 **JSPS Science Dialogue**, Yamanashi-shi, Japan.
Outreach talk on quantum physics oriented to Japanese High-School students.
Talk title: *The Applications of Quantum Physics*.
- 2016 **European Researcher's Night** Madrid
Outreach activity oriented to the communication of basic features of quantum mechanics to young children. Activity title: *Contando la cuántica (Explaining quantum physics)*
-

SCIENTIFIC RESPONSABILITIES AND COMPETENCES

- I have attracted and managed funding from three highly competitive schemes (JSPS, Marie Curie IF, and La Caixa Junior Leader), **managing a total research budget of more than 0.41M€**.
- I have produced **45 referee reports** (verified by Publons), consulted by high-quality journals such as Nature Communications, Physical Review Letters, Physical Review X and Quantum, among others.
- External supervisor for the **EU Quanterra Program**.
- External expert for Master's project exams at École polytechnique fédérale de Lausanne (EPFL), Switzerland.
- Sole supervisor of one PhD student (Alejandro Vivas Viaña). Assistant supervisor of PhD student Ariadna Soro Álvarez at Chalmers University of Technology, Sweden.
- Tutor for the “Beca de Colaboración” (Spanish Ministry of Science), “[2021 GEFES Student Prize](#)”, and “[Premios de Investigación Instituto Nicolás Cabrera for Physics Students](#)” programmes.
- Mentor in the [IFIMAC Mentoring Programme for Female Physics Students](#).